

KPI Catalog

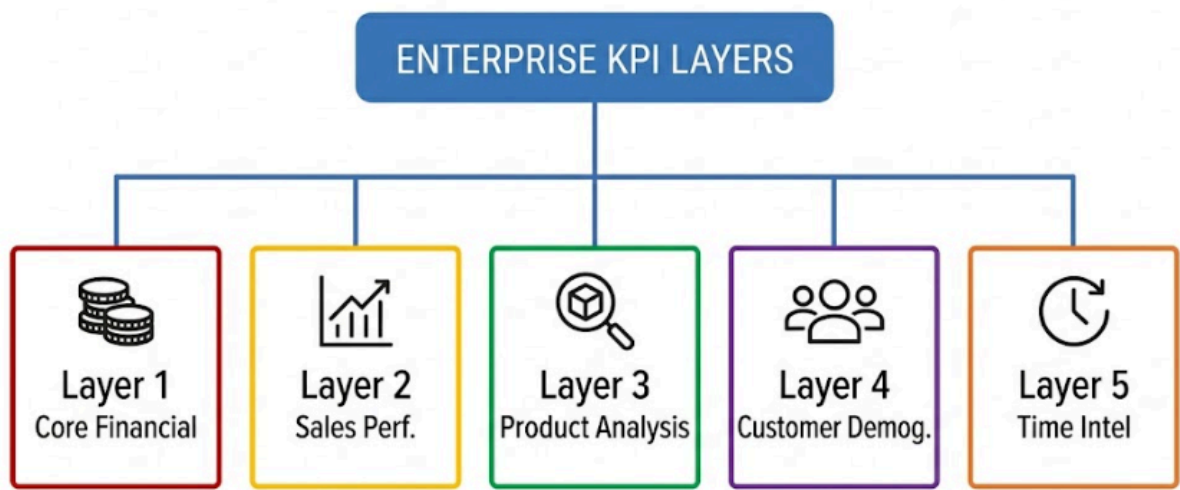
Document Metadata

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1. Document Purpose & Framework Overview

In enterprise business intelligence initiatives, the **KPI Catalog** serves as the vital bridge between high-level business definitions and physical DAX formulations. Standardizing calculations prior to report design prevents metric drift and ensures that different functional business units query a single source of truth.

To support this design, our metric framework is categorized into five cohesive layers, mapped directly to three distinct Power BI applications:



2. Platform Application Matrix

To deliver targeted analytical experiences, the enterprise metrics will be segmented across three role-based applications:

1. **Executive Overview App (Strategic):** Tailored for C-Level leadership. Focuses on overall financial health, high-level trends, and market share.

- 2. **Sales Performance App (Operational):** Designed for Sales Directors and Regional Managers. Analyzes quota fulfillment, regional targets, and individual representative contributions.
- 3. **Product & Customer Analytics App (Tactical):** Created for Product Category Managers and Marketing Analysts. Delivers deep exploratory power into buying patterns and inventory velocity.

3. Comprehensive KPI Catalog

Layer 1: Core Financial Metrics (Strategic Baseline)

KPI ID	KPI Name	Business Definition	Mathematical Formulation	Primary Target App
FIN-01	Total Revenue	Total gross revenue generated across all B2C and B2B channels before tax.	$R_{total} = R_{internet} + R_{reseller}$	Executive Overview
FIN-02	Total Product Cost	Total Cost of Goods Sold (COGS), incorporating all production or acquisition charges.	$C_{total} = C_{internet} + C_{reseller}$	Executive Overview
FIN-03	Total Gross Profit	Absolute dollar value retained after direct product costs are deducted from total revenue.	$P_{total} = R_{total} - C_{total}$	Executive Overview
FIN-04	Gross Profit Margin %	Percentage of total revenue that translates directly into gross profit.	$GP\% = \left(\frac{P_{total}}{R_{total}} \right) \times 100$	Executive Overview

FIN-05	Total Sales Units	Sum of physical units sold across both commercial transaction channels.	$U_{total} = \sum (OrderQuantity)$	Executive Overview
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Layer 2: Sales Performance & Channel Metrics (Operational Execution)

KPI ID	KPI Name	Business Definition	Mathematical Formulation	Primary Target App
SAL-01	Internet Sales Revenue	Total gross revenue originating from direct B2C e-commerce orders.	$R_{internet} = \sum (FactInternetSales(SalesAmount))$	Sales Performance
SAL-02	Reseller Sales Revenue	Total gross revenue generated via bulk B2B distribution orders.	$R_{reseller} = \sum (FactResellerSales(SalesAmount))$	Sales Performance
SAL-03	Channel Contribution %	Percentage share of enterprise revenue generated by a specific channel (i).	$CC_i\% = \left(\frac{R_{channel_i}}{R_{total}} \right) \times 100$	Sales Performance
SAL-04	Sales Quota	Aggregated target revenue milestones assigned to active sales	$Q_{total} = \sum (FactSalesQuota(SalesAmountQuota))$	Sales Performance

		employees.		
SAL-05	Quota Achievement %	The percentage of the allocated quota that was successfully realized by active representatives.	$QA\% = \left(\frac{R_{reseller}}{Q_{total}} \right) \times 100$	Sales Performance
SAL-06	Variance to Target	The absolute dollar difference between actual reseller sales and established quotas.	$V = R_{reseller} - Q_{total}$	Sales Performance
SAL-07	Average Order Value (AOV)	The average revenue generated per transaction across selected periods.	$AOV = \frac{Revenue}{Count(SalesOrderNumbers)}$	Sales Performance

Note: Since only internal Sales Representatives manage corporate B2B Reseller accounts, Quota Achievement (SAL-05) and Variance to Target (SAL-06) directly match Reseller Revenue against the assigned quotas.

Layer 3: Product Performance Metrics (Tactical Inventory)

KPI ID	KPI Name	Business Definition	Mathematical Formulation	Primary Target App
PRO-01	Product Revenue	Total gross revenue generated by individual SKUs	$R_{product} = \sum (SalesAmount)$	Product & Customer

		or design models.		
PRO-02	Product Profit	Dollar margin achieved for individual SKUs after deducting specific manufacturing cost.	$P_{\text{product}} = R_{\text{product}} - C_{\text{product}}$	Product & Customer
PRO-03	Product Contribution %	The percentage revenue contribution of a specific product (P) relative to overall sales.	$PC\% = \left(\frac{R_{\text{product}_p}}{R_{\text{total}}} \right) \times 100$	Product & Customer
PRO-04	Category Revenue	Gross revenue consolidated across high-level categories (e.g., Bikes, Accessories).	$R_{\text{cat}} = \sum_{p \in \text{Category}} R_{\text{product}_p}$	Product & Customer
PRO-05	Category YoY Growth %	Rate of category-level revenue progression compared to the same prior year period.	$CG\% = \left(\frac{R_{\text{cat, Cur}} - R_{\text{cat, LY}}}{R_{\text{cat, LY}}} \right) \times 1$	Product & Customer

Layer 4: Customer Analytics & Demographic Metrics (Tactical Marketing)

KPI ID	KPI Name	Business Definition	Mathematical Formulation	Primary Target App
CUS-01	Active Customer Count	Total unique B2C customers who completed transactions within the analyzed timeframe.	$N_{\text{cust}} = \text{DistinctCount}(\text{CustomerKey})$	Product & Customer
CUS-02	Avg Revenue per Customer	Mean gross spend per active B2C shopper.	$ARPU = \frac{R_{\text{internet}}}{N_{\text{cust}}}$	Product & Customer
CUS-03	Demographic Group Rev	Total revenue segmented by customer demographic attributes (e.g., Income Group).	$R_{\text{demog}} = \sum_{\text{Shoppers} \in \text{Cohort}} R_{\text{internet}}$	Product & Customer

Layer 5: Time Intelligence Metrics (Standard Across All Apps)

These unified time calculations apply dynamically to core revenue indicators across all three presentation apps:

KPI ID	KPI Name	Business Definition	Mathematical Formulation	App Mapping
TIM-01	Year-to-Date Revenue	Cumulative revenue generated from the start of the current calendar/fiscal year to date.	$R_{\text{YTD}} = \int_{\text{Jan 1}}^{\text{DateCur}} R(t) dt$	All Apps

TIM-02	Month-to-Dat e Revenue	Cumulative revenue generated from the start of the current month to date.	$R_{\text{MTD}} = \int_{\text{Month Start}}^{\text{Date}_{\text{cur}}} R(t) dt$	All Apps
TIM-03	Prior Year Revenue (LY)	Revenue achieved during the same historical calendar period in the previous year.	$R_{\text{LY}} = R(\text{Period}_{\text{current}} - 1 \text{ Year})$	All Apps
TIM-04	Year-over-Yea r Growth %	Percentage change in performance relative to the same period of the prior calendar year.	$\text{YoY Growth \%} = \left(\frac{R_{\text{total}} - R_{\text{LY}}}{R_{\text{LY}}} \right) \times 100$	All Apps
TIM-05	Month-over-Month Growth %	Percentage change in performance relative to the immediate previous calendar month.	$\text{MoM Growth \%} = \left(\frac{R_{\text{total}} - R_{\text{LM}}}{R_{\text{LM}}} \right) \times 100$	All Apps
TIM-06	Rolling 12-Month Revenue	Cumulative revenue generated across the previous 12 continuous calendar months.	$R_{\text{R12M}} = \int_{\text{Date}_{\text{cur}} - 12 \text{ Months}}^{\text{Date}_{\text{cur}}} R(t) dt$	All Apps

4. Architectural Implementation Best Practices

To safeguard report performance and logical consistency, BI developers must adhere to the following model practices during implementation:

1. **Separation of Base Measures:** Always instantiate simple base aggregations (e.g., [Sum of SalesAmount]) as hidden measures, then reference them inside complex business formulas (e.g., [Total Revenue]).
2. **Safe Division Handlers:** Always use the DAX DIVIDE(Numerator, Denominator, AlternateResult) function instead of standard arithmetic slashes (/) to prevent divide-by-zero errors.
3. **Dedicated Measures Folder:** All measures must be organized in dedicated, non-physical folders (e.g., _KPI Measures) sorted by layer rather than left loose within SQL table schemas.
4. **Calendar Dependency:** Time-intelligence measures (TIM-01 to TIM-06) must exclusively reference the conformed, contiguous DimDate dimension rather than transaction dates in the fact tables.

5. Metric Maintenance & Version Control

As AdventureWorks expands, this catalog will scale to incorporate new analytic domains:

- **Version 2.0 :** Introduction of Inventory Performance indicators (e.g., INV-01 Stock Turnover Ratio, INV-02 Stockout Risk Score).
- **Version 3.0 :** Launch of advanced predictive and pricing metrics (e.g., PRD-01 Demand Elasticity Score, PRD-02 Projected Margin Forecasts).